

Malware Taxonomy

The family tree

Type	Definition	Spread / role
Virus	Replicates by attaching to files/programs	Requires a host file and human action
Worm	Self-propagating standalone program	Exploits network + autospreads (Code Red, Morris)
Trojan	Disguised as legitimate; isn't	Backdoors, RATs, fake updaters
Ransomware	Encrypts/locks data for payment	Often worm-ish or phishing-delivered
Spyware	Steals info/behavior data	Bundled installs, keyloggers
Bot	Joins a C2 botnet	DDoS, spam, proxying, credential harvesting
RAT	Remote Access Tool (abused)	Full remote control, screenshots, keylog
Adware/PUP	Unwanted but 'gray'	Revenue, system hooks

The lifecycle of an infection

Delivery (phishing, drive-by, malicious USB, supply chain)

Execution (user double-click, exploit, macro, service)

Persistence – survive reboot (run keys, services)

Defense evasion – obfuscate, pack, polymorph

C2 – beacon home

Actions – exfil, ransom, destroy

Malware needs

Armoring: packers (UPX), cryptors, obfuscators, polymorphism – so AV signatures miss

Persistence + evasion: living off the land (PowerShell/mshta/cscript) to blend in

Kill-chains in time: stay quiet (dwell time), beacon low-and-slow

Modern realism

Most malware today is **loaders + C2**: the first-stage is small; the final payload downloads later

Ransomware *operates* (EDR on) – defensive success = stop chain early

APTs weaponize **dual-use** tools (PowerShell, certutil, BITS) so organic behavior fits

What an analyst watches

Event sources: process creation, network connections, registry writes, file writes, scheduled tasks, WMI, services, autoruns, child processes.

The team that knows malware generics (symptom, spread, survival) already sees the attack half-clear before the sandbox report lands.

[← Module start](#)

[Ransomware: Operators, Tactics...](#)